

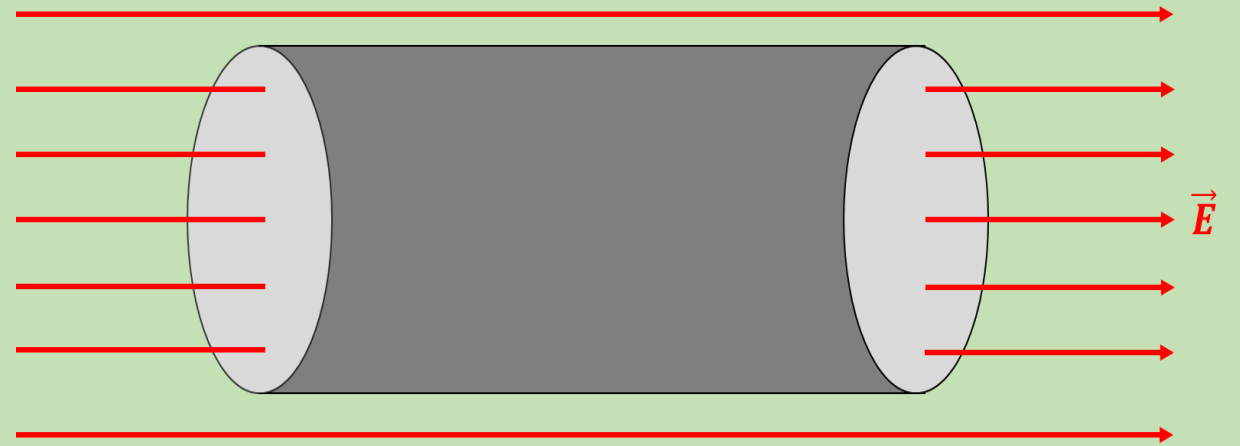
Lecture 9

Chapter 23

Gauss' Law

Problems Solved Class

Q1. Suppose a hypothetical cylinder of radius R is immersed in a uniform electric field $\vec{E}$ and the cylinder axis is parallel to the field. Find the net electric flux for this closed surface.



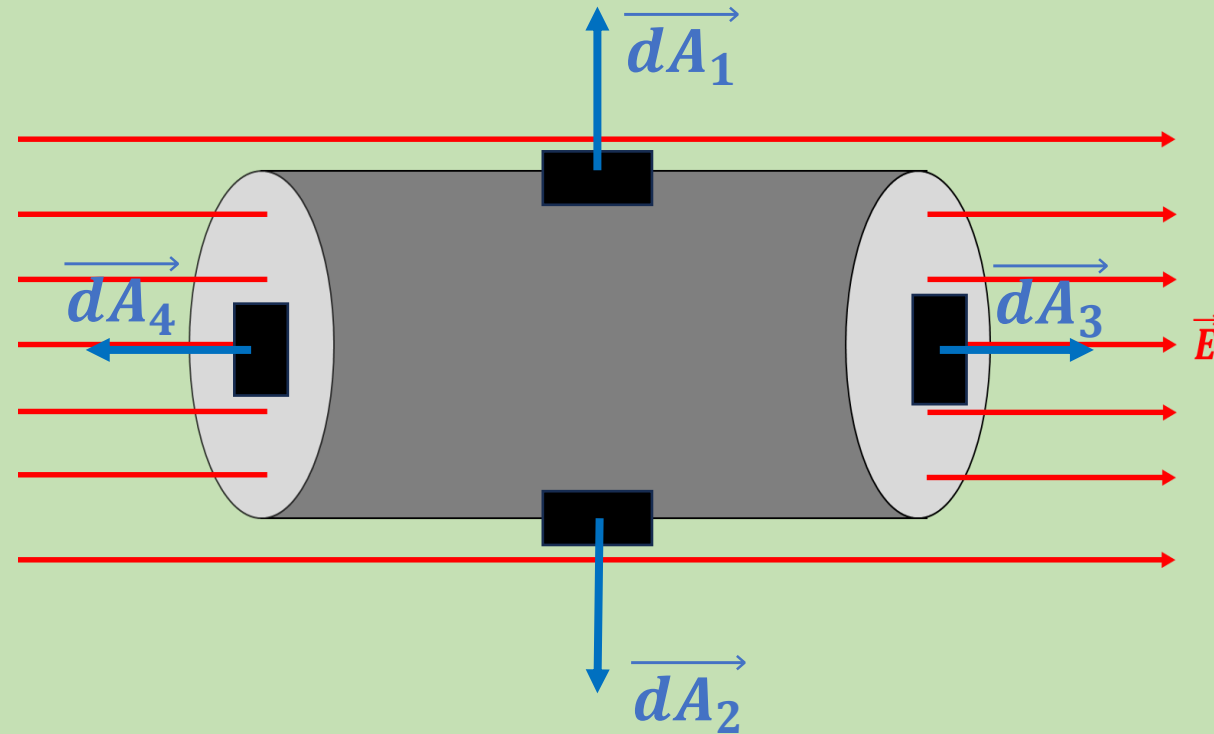
Ans:

Let, the infinitesimal surface $\vec{dA_1}$ and $\vec{dA_2}$ on the curved region, and the infinitesimal surface $\vec{dA_3}$ and $\vec{dA_4}$ on the circular region.

Flux for the curved surface,

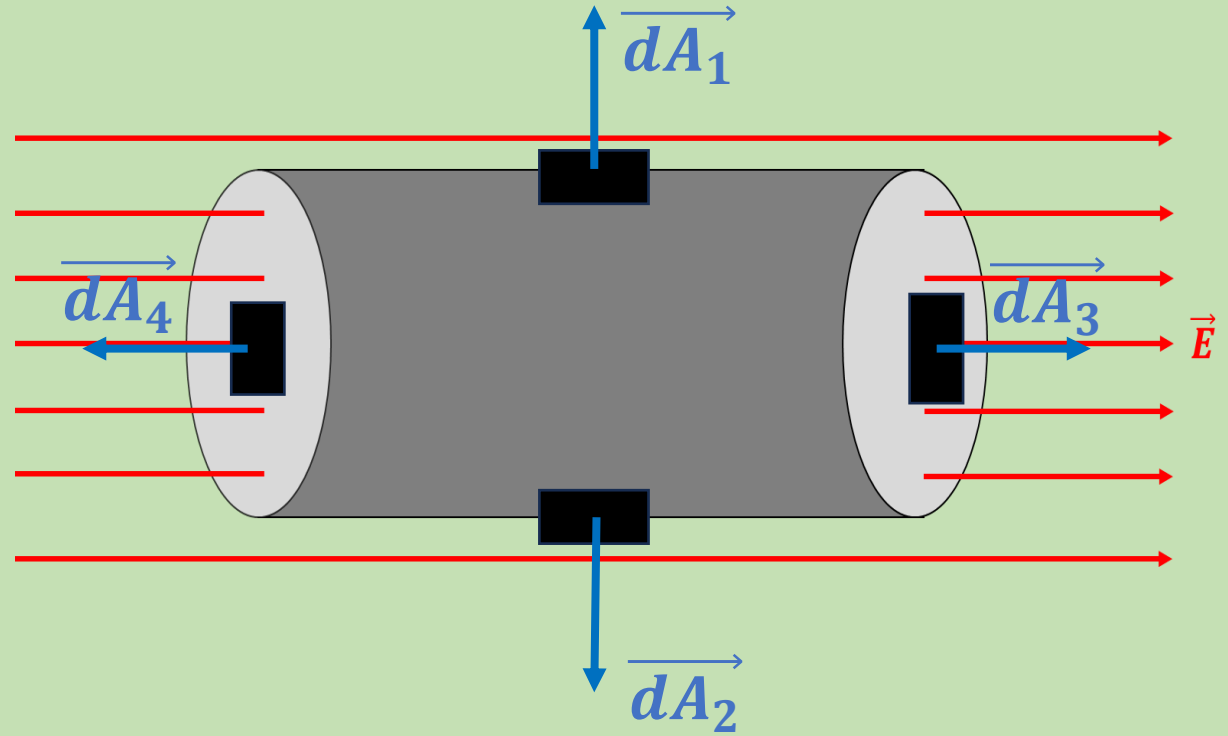
$$\Phi_1 = \oint \vec{E} \cdot \vec{dA_1} = \oint E dA_1 \cos 90^\circ = 0$$

$$\Phi_2 = \oint \vec{E} \cdot \vec{dA_2} = \oint E dA_2 \cos 90^\circ = 0$$



Flux for the circular surface,

$$\begin{aligned}\Phi_3 &= \oint \vec{E} \cdot \vec{dA}_3 \\&= \oint E dA_3 \cos 0^\circ \\&= E \oint dA_3 \\&= E\pi R^2 \\ \Phi_4 &= \oint \vec{E} \cdot \vec{dA}_4 \\&= \oint E dA_4 \cos 180^\circ \\&= E \oint dA_4 \\&= -E\pi R^2\end{aligned}$$



Therefore, net flux for the closed surface,

$$\begin{aligned}\Phi_{net} &= \Phi_1 + \Phi_2 + \Phi_3 + \Phi_4 \\&= 0 + 0 + E\pi R^2 - E\pi R^2 \\&= 0\end{aligned}$$

Q2. Suppose a hypothetical cube of each edge a is immersed in a uniform electric field $\vec{E}$ and the field is directed to the z -axis. Find the net electric flux for this closed surface.

Ans:

Let, the infinitesimal surfaces $\vec{dA_1}$, $\vec{dA_2}$, $\vec{dA_3}$, $\vec{dA_4}$, $\vec{dA_5}$ and $\vec{dA_6}$ on the face of the cube.

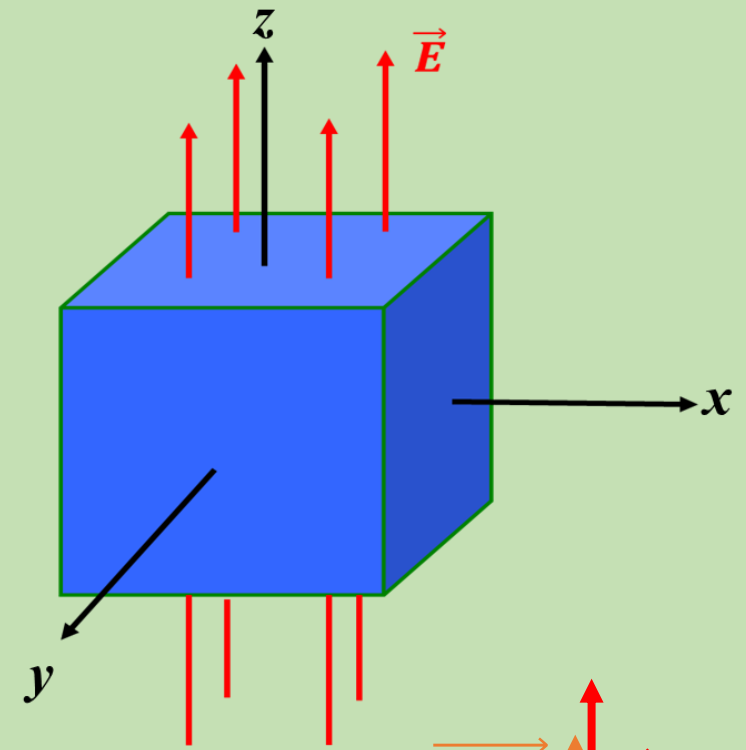
Flux for the surfaces are,

$$\begin{aligned}\Phi_1 &= \oint \vec{E} \cdot \vec{dA_1} = \oint E dA_1 \cos 0^\circ = E \oint dA_1 \\ &= Ea^2\end{aligned}$$

$$\begin{aligned}\Phi_2 &= \oint \vec{E} \cdot \vec{dA_2} = \oint E dA_2 \cos 180^\circ \\ &= E \oint dA_2 = -Ea^2\end{aligned}$$

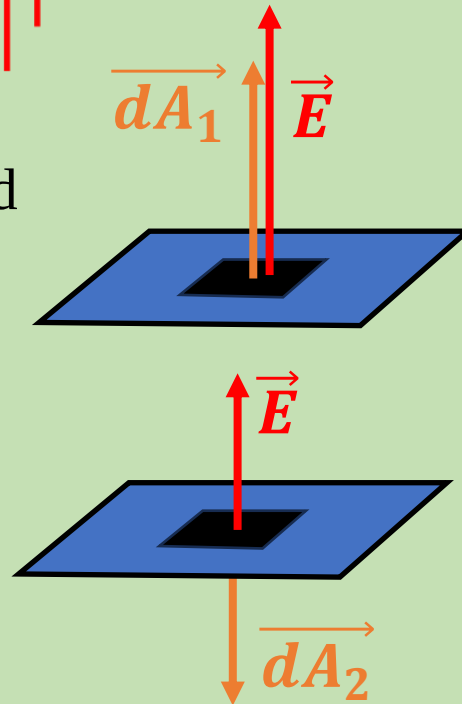
$$\Phi_3 = \Phi_4 = \Phi_5 = \Phi_6 = 0$$

(because the area vector of these surfaces is perpendicular to the electric field)

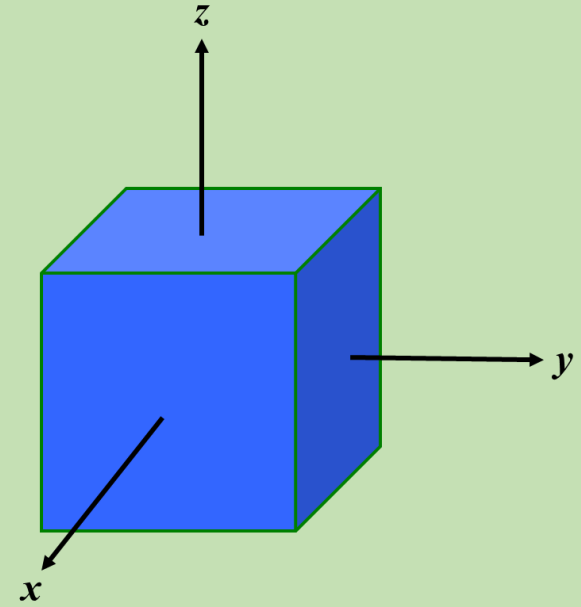


Therefore, the net flux for the closed surface,

$$\begin{aligned}\Phi_{net} &= \Phi_1 + \Phi_2 + \Phi_3 + \Phi_4 \\ &= Ea^2 - Ea^2 + 0 + 0 + 0 + 0 \\ &= 0\end{aligned}$$



Q3. Figure shows a Gaussian surface in the shape of a cube with edge length **1.40 m**. What are (a) the net flux through the surface and (b) the net charge enclosed by the surface if $\vec{E} = (3.00y\hat{j}) \text{ NC}^{-1}$ with y in meters? What are (c) the net flux and (d) the net enclosed charge if $\vec{E} = [-4.00\hat{i} + (6.00 + 3.00y)\hat{j}] \text{ NC}^{-1}$?



Ans:

(a) The net flux for the closed surface, $\Phi_{net} = 0$

(b) From the Gauss' law, $\Phi_{net} = \frac{q_{enc}}{\epsilon_0}$; So, the net charge enclosed by the surface, $q_{enc} = 0$

(c) $\vec{A}_1 = a^2\hat{i}$, $\vec{A}_2 = -a^2\hat{i}$, $\vec{A}_3 = a^2\hat{j}$, $\vec{A}_4 = -a^2\hat{j}$, $\vec{A}_5 = a^2\hat{k}$, and $\vec{A}_6 = -a^2\hat{k}$
where, $a^2 = 1.96 \text{ m}^2$

The net electric field

$$\begin{aligned}\Phi_{net} &= \vec{E} \cdot \vec{A}_1 + \vec{E} \cdot \vec{A}_2 + \vec{E} \cdot \vec{A}_3 + \vec{E} \cdot \vec{A}_4 + \vec{E} \cdot \vec{A}_5 + \vec{E} \cdot \vec{A}_6 \\ &= -4a^2 + 4a^2 + (6.00 + 3.00y)a^2 - (6.00 + 3.00y)a^2 + 0 + 0 \\ &= 0\end{aligned}$$

(d) From the Gauss' law, $\Phi_{net} = \frac{q_{enc}}{\epsilon_0}$; So, the net charge enclosed by the surface, $q_{enc} = 0$

Q4. The figure shows a closed Gaussian surface in the shape of a cube of edge length **2.00 m**, with one corner at $\mathbf{x_1 = 5.00\ m, y_1 = 4.00\ m}$. The cube lies in a region where the electric field vector is given by

$$\vec{E} = -3.00\hat{i} - 4.00y^2\hat{j} - 3.00\hat{k}\text{ NC}^{-1}$$

with $\mathbf{y}$ in meters. What is the net charge contained by the cube?

Ans:

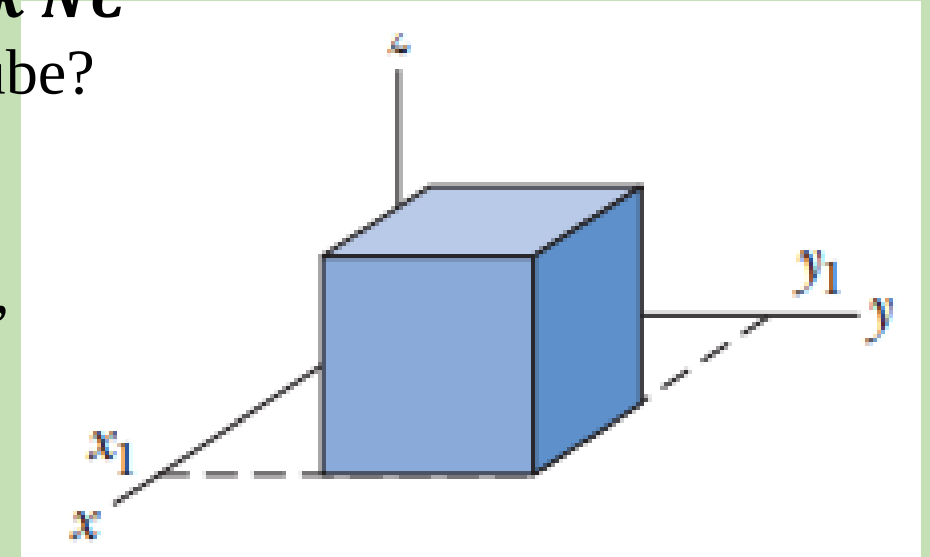
$$\vec{A_1} = a^2\hat{i}, \vec{A_2} = -a^2\hat{i}, \vec{A_3} = a^2\hat{j}, \vec{A_4} = -a^2\hat{j}, \vec{A_5} = a^2\hat{k},$$

and $\vec{A_6} = -a^2\hat{k}$; where, $a^2 = 4\text{ m}^2$

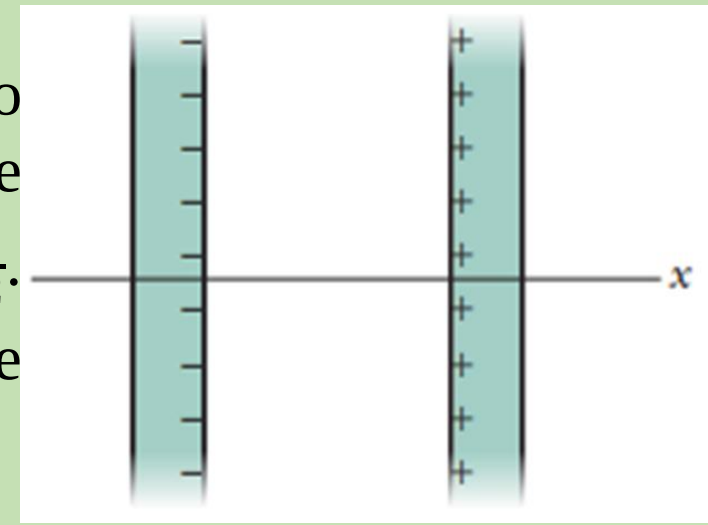
The net electric field

$$\begin{aligned}\Phi_{net} &= \vec{E} \cdot \vec{A_1} + \vec{E} \cdot \vec{A_2} + \vec{E} \cdot \vec{A_3} + \vec{E} \cdot \vec{A_4} + \vec{E} \cdot \vec{A_5} + \vec{E} \cdot \vec{A_6} \\ &= -3a^2 + 3a^2 - 4y^2a^2 + 4y^2a^2 - 3a^2 + 3a^2 \\ &= 0\end{aligned}$$

From the Gauss' law, $\Phi_{net} = \frac{q_{enc}}{\epsilon_0}$; So, the net charge enclosed by the surface, $q_{enc} = 0$



Q5. In figure, two large, thin metal plates are parallel and close to each other. On their inner faces, the plates have excess surface charge densities of opposite signs and magnitude $7.00 \times 10^{-22} \frac{C}{m^2}$. In unit-vector notation, what is the electric field at points (a) to the left of the plates, (b) to the right of them, and (c) between them?



Ans:

Let the electric field for the negative charge and the positive charge of the thin metal plates are

$$E_- = \frac{\sigma}{\epsilon_0}, \text{ and } E_+ = \frac{\sigma}{\epsilon_0}$$

Where, $\sigma = \text{surface charge density of the plates}$

(a) Left side of the plates:

$$E_{net} = 0$$

(b) Right side of the plates:

$$E_{net} = 0$$

(c) Between the plates:

$$E_{net} = E_+ + E_-$$

$$\begin{aligned} E_{net} &= E_+ + E_- \\ &= \frac{2\sigma}{\epsilon_0} \end{aligned}$$

$$\begin{aligned} &= \frac{2 \times 7 \times 10^{-22}}{8.854 \times 10^{-12}} \\ &= 1.58 \times 10^{-10} NC^{-1} \end{aligned}$$

$$\vec{E}_{net} = -(1.58 \times 10^{-10} NC^{-1})\hat{i}$$

Thank You